Reducing Hallucinations in Enterprise Customer Service LLMs via Citation-Driven Reinforcement Learning

1. Research Questions and Objectives

As large language models (LLMs) are increasingly deployed in enterprise customer service systems, ensuring that responses are verifiable and evidence-based has become critical to user trust and regulatory compliance. Although retrieval-augmented generation (RAG) introduces external knowledge sources to reduce hallucinations, models still produce unsupported claims or misuse retrieved evidence [1].

This research asks: How can reinforcement learning be used to train retrieval-augmented LLMs to consistently cite correct evidence when it exists and to refuse or seek clarification when evidence is insufficient?

The objectives are threefold: (1) to define hallucination operationally as factual claims unsupported by retrieved evidence or inconsistent with cited sources; (2) to design a reinforcement learning framework where citation accuracy, faithfulness, and appropriate refusal behavior are explicit optimization signals; and (3) to evaluate trade-offs among answer usefulness, refusal rates, user satisfaction, and operational cost.

2. Literature Review

Retrieval-augmented generation combines dense retrieval with sequence-to-sequence generation, improving factual accuracy and enabling traceable answers by conditioning responses on retrieved documents [1][4]. However, prior work shows that RAG systems may still overgeneralize or hallucinate when retrieved evidence is incomplete or noisy.

Reinforcement learning from human feedback (RLHF) has been shown to improve factuality and alignment by training models to follow human preferences, as demonstrated in browser-assisted question answering systems such as WebGPT [2]. Studies on faithful reasoning further argue that models must align generated conclusions with intermediate evidence and reasoning steps to ensure verifiability [3].

Despite these advances, existing approaches rarely optimize for decision-level behaviors—such as when to refuse answering or request clarification—especially in enterprise customer service contexts where risk sensitivity is essential. This gap motivates a reinforcement learning approach that explicitly encodes citation correctness and uncertainty handling.

3. Methodology

We propose a system-level framework consisting of a retriever, a generator, and a policy learner. The policy learner selects among three actions: generating a cited answer, asking a clarification question, or refusing to answer.

Reinforcement learning is used to optimize this policy with a composite reward function: positive reward for citation accuracy and evidence-faithful answers; penalties for unsupported claims or incorrect citations; and calibrated rewards for refusal or clarification when evidence coverage is insufficient. Preference-based learning is incorporated to encode a risk-averse bias that favors refusal over hallucination, consistent with prior instruction-following alignment methods [5].

Training and evaluation will use enterprise QA datasets with annotated evidence, combined with simulated ambiguous queries. Metrics include hallucination rate, citation precision, refusal appropriateness, user satisfaction proxies, and average handling time.

4. Expected Outcomes and Significance

We expect the proposed method to significantly reduce hallucinations and citation errors while moderately increasing appropriate refusals and clarification requests. Over multi-turn interactions, clarification strategies are expected to improve evidence coverage and reduce high-risk incorrect answers. This research contributes a practical and auditable framework for deploying LLM-based customer service systems in high-stakes enterprise environments.

References

[1] Lewis P, Perez E, Piktus A, et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS, 2020. *https://arxiv.org/abs/2005.11401*

[2] Nakano R, Hilton J, Balaji S, et al. WebGPT: Browser-assisted Question-Answering with Human Feedback. arXiv:2112.09332, 2021. *https://arxiv.org/abs/2112.09332*

[3] Creswell A, Shanahan M. Faithful Reasoning Using Large Language Models. arXiv:2208.14271, 2022. *https://arxiv.org/abs/2208.14271*

[4] Wikipedia. Retrieval-augmented generation. *https://en.wikipedia.org/wiki/Retrieval-augmented_generation*

[5] Ouyang L, Wu J, Jiang X, et al. Training Language Models to Follow Instructions with Human Feedback. NeurIPS, 2022. *https://arxiv.org/abs/2203.02155*